

Effect of timely manoeuvre execution on the collision probability in head-on and crossing encounter scenarios

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ABSTRACT: This paper assesses the collision probability in different encounter scenarios considering the manoeuvring capabilities of the vessels. The collision risk is analysed by trajectory simulation of the two encounter ships in several scenarios: “blind” navigation, COLREG execution (i.e., correct following of the rules by both vessels), and COLREG-blind (i.e., only one of the vessels follows the rules). The trajectories of the ships involved in collision scenarios are generated probabilistically by Monte Carlo simulation using the first-order Nomoto manoeuvrability model. This paper provides some indications for the relative distance between ships at which a collision avoidance manoeuvre should be executed to reduce the collision probability using the minimum safe distance concept. The generic simulation model based on simple input data and manoeuvring characteristics of the vessels can be later improved for collision avoidance applications.

1 INTRODUCTION

Ship collision is a navigational accident that accounts for 13% of all maritime casualties in the EU between 2014 and 2020 (EMSA, 2021). Moreover, ship collisions are responsible for 32% of fatalities in maritime accidents over the last six years. Globally, the situation is not much different, with collisions being the second most frequent cause of incidents (after machinery damage/failure) according to Allianz (2022). Therefore, ship collision assessment is still an important research topic, as evidenced by recent studies (e.g. Antão et al., 2023; Rong et al., 2022; Silveira et al., 2022; Uyanik et al., 2021).

Collision assessment can encompass various operational scenarios, as demonstrated by previous research studies. Montewka et al. (2012) analysed ship-ship collision accidents, while (Pedersen, 2014; Yu et al., 2020) focused on collision scenarios involving ships and offshore structures. Thieme et al. (2018) concentrated on the collision risk assessment of marine autonomous surface ships.

The assessment of collision risk on board a vessel is typically carried out by the officers responsible for navigation watch using the radar and the Automatic Radar Plotting Aid (ARPA) technology (Krata & Montewka, 2015). This technology monitors nearby vessels and displays the Closest Point of Approach (CPA), which is the minimum distance at which two ships will pass each other, and the Time to Closest Point of Approach (TCPA). In situations requiring collision avoidance manoeuvres, mariners refer to the International Regulations for Preventing Collision at Sea (COLREG). However, in cases where immediate response is necessary or the relative velocity is high,

CPA and TCPA become less significant (Silveira et al., 2016).

Unfortunately, the correct following of COLREG rules does not necessarily guarantee collision avoidance. Due to different interpretations of collision risk, it is impractical to incorporate all of them and establish a single definition (Silveira et al., 2021). Consequently, the COLREG's definition of collision risk is subjective and dependent on the mariner's experience and judgement (Silveira et al., 2016). For instance, two officers on watch may interpret the same situation differently and arrive at a close-quarter encounter despite applying the same set of rules (Kuwata et al., 2014).

An additional ambiguity issue in the COLREG regulation concerns the moment when a ship must take measures to avoid a collision (Silveira et al., 2021). For instance, it is uncertain at which point a ship that was not initially obligated to take action to avoid a collision should do so if the other vessel fails to take action or takes insufficient action.

The ship manoeuvrability is a crucial element in collision probability studies that aim to improve collision risk assessment. Curtis (1986) investigated ship dynamics to determine the safe distance for a Very Large Crude Carrier's (VLCC) overtaking manoeuvre. Krata & Montewka (2015) utilised a quasi-linear modular hydrodynamic model, developed by Matusiak (2011), to estimate the critical area for a ship involved in a collision encounter. Silveira et al. (2016) analysed evasive manoeuvring movements in near-collision scenarios using AIS data and the first-order Nomoto model. Szlapczynski et al. (2018) integrated ship manoeuvrability and ship domain to identify the latest possible time for conducting a collision avoidance manoeuvre safely.

Based on hydrodynamic and manoeuvrability characteristics, Montewka et al. (2010) introduced the concept of minimum distance to collision (MDTC) – the minimum distance between two ships on collision courses at which successful evasive manoeuvres can be performed.

The objective of this paper is to present a preliminary collision risk assessment approach based on simple input data that can be further developed for collision avoidance applications. The model evaluates the risk of collision in head-on and crossing encounters, taking into account the manoeuvring capabilities of the vessels, which are modelled using Nomoto's first-order equation. The concept of minimum safe distance is introduced and used to assess the probability of collision between two ships in three scenarios: COLREG-COLREG, COLREG-blind, and blind-blind.

2 METHODOLOGY

2.1 Minimum safe distance

In this study, the probability of collision between two vessels in a close-quarter situation is assessed using the concept of minimum safe distance. Minimum safe distance refers to the minimum distance between two vessels at which collision does not occur, considering the vessels' manoeuvrability characteristics, initial speed and rudder angle, main dimensions, and vessel type. The minimum safe distance is determined for a particular set of initial conditions, depending on the type of encounter (i.e. head-on or crossing) and scenario (i.e. COLREG-COLREG, COLREG-blind, blind-blind). Figure 1 presents the top-down diagram with the elements of the implemented procedure.

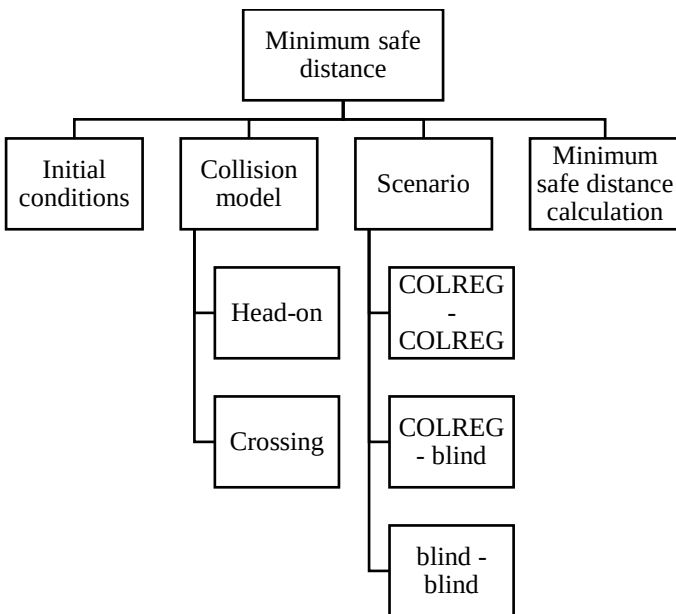


Figure 1. Top-down diagram of the implemented model.

The assessment of the minimum safe distance involves a while loop algorithm that iteratively establishes the turning circle manoeuvre of both vessels (with a particular velocity and rudder angle) with increasing initial relative distance until the collision between the vessels does not occur.

Prior to the start of the while loop, both vessels are positioned in a bow-to-bow configuration regardless of the encounter (see Figure 2). The distance between the vessels is then iteratively incremented by 50 m until no overlap is observed. At this point the algorithm stops and the initial distance between the ships is registered as the minimum safe distance.

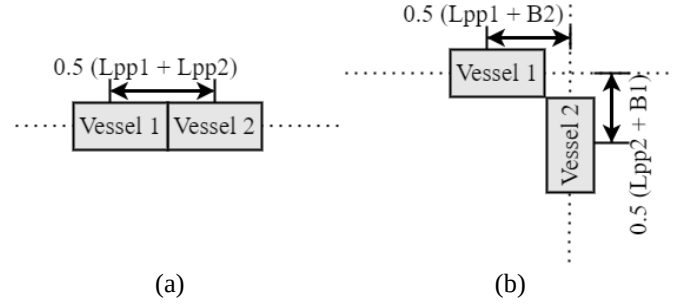


Figure 2. Initial position of the vessels in head-on (a) and crossing (b) encounters

2.2 Collision probability assessment

The probability of collision is defined as the probability that the relative distance between two vessels, d , is lower than the minimum safe distance, d_{min} , estimated by:

$$P_{collision}(d) = P(d_{min} > d) = 1 - F_{d_{min}}(d) \quad (1)$$

where the probability distribution function, $F_{d_{min}}$, of the minimum safe distance, d_{min} , is obtained by Monte Carlo simulation of the minimum safe distance assessment procedure just described. The resulting sample of size n of minimum safe distances account for the uncertainties on the initial speed and rudder angle of the ships (i.e. uncertainties on the initial conditions of the ships in the encounter). Based on this dataset the probability density function $f_{d_{min}}(d)$ can be established and the cumulative distribution function of the minimum safe distance is then:

$$F_{d_{min}}(d) = P(d_{min} \leq d) = \int_{-\infty}^d f_{d_{min}}(d) dd, \quad \forall d \in \mathbb{R} \quad (2)$$

The accuracy of the results provided by MCS is influenced by the number of simulation histories used (Lotovskyi et al., 2022). To determine the appropriate number of histories, the error related to the Confidence Interval (CI) can be estimated using Equation 3.

$$Error_{CI\ 95\%} = \frac{1.96 \frac{\sigma}{\sqrt{n}}}{\mu} \times 100 \quad (3)$$

where σ = standard deviation; n = sample size; and μ = sample mean.

2.3 Nomoto's model

To apply the proposed methodology and evaluate the resulting outcomes, it is necessary to mathematically define the manoeuvring behaviour of a vessel. For this purpose, the first-order Nomoto model is employed (Nomoto et al., 1957). While this model may not be the most precise method to predict the vessel's trajectory during manoeuvring (Sutulo & Guedes Soares, 2011), it is a straightforward model to implement and does not require any specialised information, such as data from hydrodynamic experiments or zigzag sea trial tests in open waters. The easily accessible data (such as AIS data) can be used to identify the main dimensions and vessel type, which are sufficient for implementing the manoeuvre in this case study (Silveira et al., 2016).

The first order Nomoto model in time domain can be expressed by Equation 4 (Nomoto et al., 1957).

$$T\dot{r} + r = K\delta \quad (4)$$

where K = static yaw rate gain; T = governance time constant; r = ROT; and δ = rudder angle.

Besides, the model can be defined by Equation 5.

$$T\ddot{\psi} + \dot{\psi} = K\delta, \quad \dot{\psi} = r \quad (5)$$

where, ψ = heading angle of the ship that can be defined by Equation 6.

$$\psi_{i+1} = \psi_i + r_{i+1}(t_{i+1} - t_i), \quad i \in \mathbb{N} \quad (6)$$

By solving Equation 4, ROT can be determined by Equation 7.

$$r = \frac{r_0}{e^{t/T}} + \delta K \left(1 - \frac{1}{e^{t/T}}\right) \quad (7)$$

where, r_0 = initial ROT.

An initial approximation of the parameter T can be obtained by considering the vessel's length and type. For this study, the triangular probability distribution suggested by Silveira et al. (2016) is adopted to model the T parameter (see Table 1).

Static yaw rate gain, K , can be estimated by Equation 8.

$$K = \frac{V_{STD}^{(35)}}{\delta R}, \quad \text{where } R = 2 \times Lpp \quad (8)$$

where R = steady turning radius (SNAME, 1989); and $V_{STD}^{(35)}$ = steady turn speed for the maximum rudder angle defined in Equation 9.

$$V_{STD}^{(35)} = V_0 \times \frac{V_{ST}}{V_0}, \quad \text{where } \frac{V_{ST}}{V_0} \sim U(0.6, 0.7) \quad (9)$$

where V_0 = initial speed; and V_{ST}/V_0 = ratio sampled from a uniform probability distribution for turning circles executed using maximum rudder angle (Silveira et al., 2016).

The speed of the vessel, $V^{(\delta)}$, at instant t and rudder angle δ is modelled in Equation 10.

$$V^{(\delta)} = V_0 \left(1 - C_{Vred}^{(\delta)} \times \frac{tg^2(\beta)}{tg^2(\beta_{STD})}\right), \quad (10)$$

$$\text{where } \beta_{STD} \sim U\left(18 \frac{Lpp}{R}, 22.5 \frac{Lpp}{R} + 1.45\right)$$

where $C_{Vred}^{(\delta)}$ = speed reduction coefficient for rudder angle δ (see Equations 11 and 12); β = drift angle; and β_{STD} = steady turning drift angle defined probabilistically by a uniform distribution (SNAME, 1989).

$$C_{Vred}^{(\delta)} = \frac{\delta}{35^\circ} \times C_{Vred}^{(35^\circ)} \quad (11)$$

$$C_{Vred}^{(35^\circ)} = 1 - \frac{V_{ST}}{V_0} \quad (12)$$

The effect of drift angle, β , on the turning circle manoeuvre can be estimated by Equation 13.

$$T\dot{\beta} + \beta = K_\beta \delta \quad (13)$$

Considering:

$$\beta = \beta_{ST} \Leftrightarrow \dot{\beta} = 0 \quad (14)$$

$$K_\beta = \frac{\beta_{ST}}{\delta} \quad (15)$$

From Equation 14, the drift angle at instant t can be estimated by Equation 16.

$$\beta = \frac{\beta_0}{e^{t/T}} + \delta K_\beta \left(1 - \frac{1}{e^{t/T}}\right) \quad (16)$$

The position change of pivot point is described by Equation 17.

$$\begin{cases} x_{i+1} = x_i + \Delta d \times \cos(COG) \\ y_{i+1} = y_i + \Delta d \times \sin(COG) \end{cases} \quad (17)$$

where $COG = \psi - \beta$

where Δd = sailed distance between t_i and t_{i+1} .

3 CASE STUDY DESCRIPTION

The aim of this paper is to conduct an initial assessment of the collision probability in different encounter scenarios, considering the manoeuvring capabilities of the vessels involved.

The collision probability is assessed via trajectory simulation of two cargo ships with identical length, L_{pp} , and beam, B , modelled as rectangles whose movement is estimated based on Course Over Ground (COG), Speed Over Ground (SOG), and heading data. The model assumes that a fixed pivot point is located at the midship position. This modelling approach is adopted for simplicity reasons. The simulation considers head-on and crossing scenarios in deep water and good visibility condition (see Figure 3). Weather conditions are excluded from the analysis. The initial heading, ψ_0 , is set equal to the initial COG, the initial drift angle, β_0 , and the initial Rate of Turn (ROT) are held at zero. The vessels' initial service speed, V_0 , follows a uniform distribution. A summary of the initial assumptions is presented in Table 1.

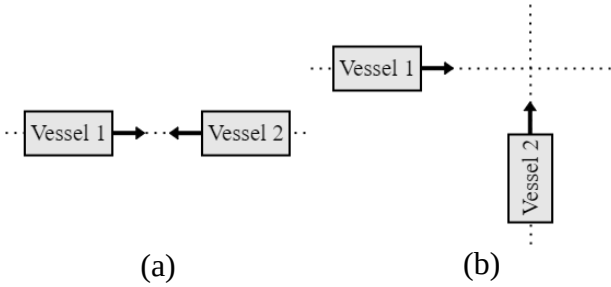


Figure 3. Collision risk assessment of head-on encounter (a) and crossing situation (b).

Three distinct scenarios are employed for both head-on and crossing collision encounters: COLREG-COLREG, COLREG-blind, and blind-blind (as shown in Table 2). It should be noted that the initial positions of each vessel are dependent on the collision encounter type, which in turn determines the specified initial heading angles as outlined in Table 2.

COLREG-COLREG corresponds to the scenario in which two vessels behave in accordance with the Collision Regulations (COLREG). When vessels are approaching each other head-on, Rule 14 of COLREG dictates: "when two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision each shall alter her course to starboard so that each shall pass on the port side of the other" (Cockcroft & Lameijer, 2012). In order to replicate this statement in the model, the probabilistic behaviour of rudder angle, δ , follows a triangular distribution ranging from 1° to 35° , with the most likely angle being 20° (see Table 2).

In a crossing situation, Rule 15 of COLREG mandates that "the vessel which has the other on her own starboard side shall keep out of the way and shall, if

the circumstances of the case admit, avoid crossing ahead of the other vessel" (Cockcroft & Lameijer, 2012). In this case, the stand-on vessel will have a fixed rudder angle of 0° , while the give-way vessel will alter her course to starboard. The rudder angle change of the give-way vessel follows a triangular distribution between 1° to 35° , with the most likely angle being 20° (see Table 2).

The scenario COLREG-blind arises when one vessel follows the COLREG regulation, while the other exhibits random behaviour. This situation may arise due to various reasons, such as the "blind" vessel misinterpreting the information from another ship or the "blind" vessel identifying the encounter situation too late. For vessels with "blind" rudder angle behaviour, a uniform distribution is applied (see Table 2).

Blind-blind scenario entails both vessels having a random angle of rudder, which may arise from a communication error between bridges or, for example, a misinterpretation of each other's actions. In this scenario, both vessels exhibit a "blind" rudder angle behaviour that follows a uniform distribution, as specified in Table 2.

Table 1. Initial input data considerations.

Variables	Data
Type of the vessel	Cargo ship
L_{pp} (m)	330
B (m)	50
V_0 (knots)	$V_0 \sim U(10, 18)$
β_0 ($^\circ$)	0
ROT ($0^\circ/s$)	0
T parameters:	
- Minimum	70
- Most likely	100
- Maximum	130

Table 2. Initial values for heading and rudder angles in different encounter scenarios.

	Encounter scenario	
	Head-on	Crossing
$\psi_0^{vessel 1}$ ($^\circ$)	90	90
$\psi_0^{vessel 2}$ ($^\circ$)	270	0
COLREG - COLREG	δ_1 ($^\circ$) $\sim$ Tri (1, 20, 35) δ_2 ($^\circ$) $\sim$ Tri (1, 20, 35)	δ_1 ($^\circ$) $\sim$ Tri (1, 20, 35) δ_2 ($^\circ$) = 0
COLREG - blind	δ_1 ($^\circ$) $\sim$ Tri (1, 20, 35) δ_2 ($^\circ$) $\sim$ U (-35, 35)	δ_1 ($^\circ$) $\sim$ Tri (1, 20, 35) δ_2 ($^\circ$) $\sim$ U (-35, 35) Or δ_1 ($^\circ$) $\sim$ U (-35, 35) δ_2 ($^\circ$) = 0
blind - blind	δ_1 ($^\circ$) $\sim$ U (-35, 35) δ_2 ($^\circ$) $\sim$ U (-35, 35)	δ_1 ($^\circ$) $\sim$ U (-35, 35) δ_2 ($^\circ$) $\sim$ U (-35, 35)

4 DISCUSSION AND RESULTS

4.1 Turning circle manoeuvre

Figure 4 shows the trajectory of the midship point of the analysed vessel during a turning manoeuvre. The simulation assumes an initial vessel speed of 15 knots and a rudder angle of 35° . Under these conditions, the resulting transfer, advance, tactical diameter, and turning radius are 735 m, 1557 m, 1540 m, and 661 m, respectively.

4.2 Number of simulation histories

Any simulation analysis starts by determining the optimal number of simulated scenarios (i.e. histories) for a given level of accuracy. Increasing the number of simulations yields more precise results. To estimate the necessary number of simulations, the error associated with the 95% Confidence Interval of the probability of collision is considered. Two distinct models are utilised for this purpose: one for the head-on encounter scenario, and another for the crossing situation.

To make this estimation, the probability of collision is calculated as the ratio of the number of collisions that occur to the total number of simulations conducted for various rudder angles and initial speeds. In each simulation, the initial distance between vessels is kept constant and the uncertainties associated with the parameters of the manoeuvrability model are taken into account.

The optimal number of simulation histories is determined individually for the head-on encounter and crossing situation models. Figure 5 displays the obtained results. Given limitations in computer processing power and the errors associated with the 95% CI, 10,000 simulation histories are chosen for both models. The resulting error for the head-on encounter model is 2.65%, while for the crossing situation it is 3.64%.

4.3 Probability of collision in head-on encounter

Considering manoeuvring characteristics of the vessel and applying the optimum number of simulation histories, this section presents a preliminary risk assessment for collisions in head-on encounters.

Figure 6 shows the histogram of the minimum safe distance, along with the corresponding probability distribution function for collision events in the COLREG-COLREG scenario. The results reveal that when both vessels adhere to the rules outlined in COLREG, the probability of collision is low for distances up to 1.1 nm between the vessels. However, for the type of vessel considered in this study, collisions are unavoidable when the distance between vessels is less than 0.85 nm, even if both vessels are following the COLREG rules.

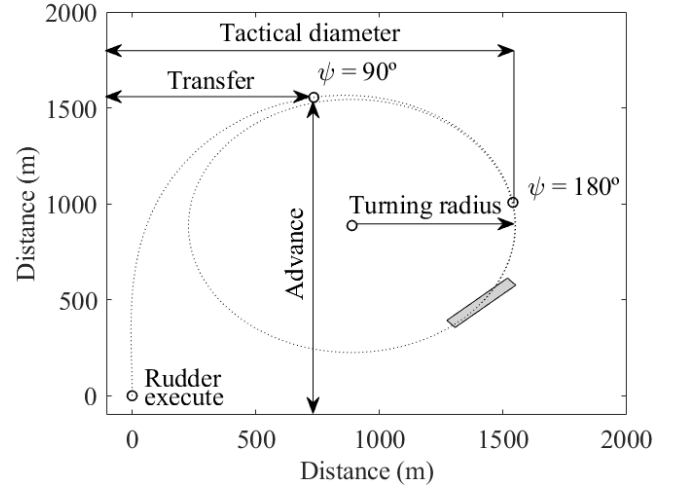


Figure 4. Path of midship point in turning circle and main dimensions of manoeuvre.

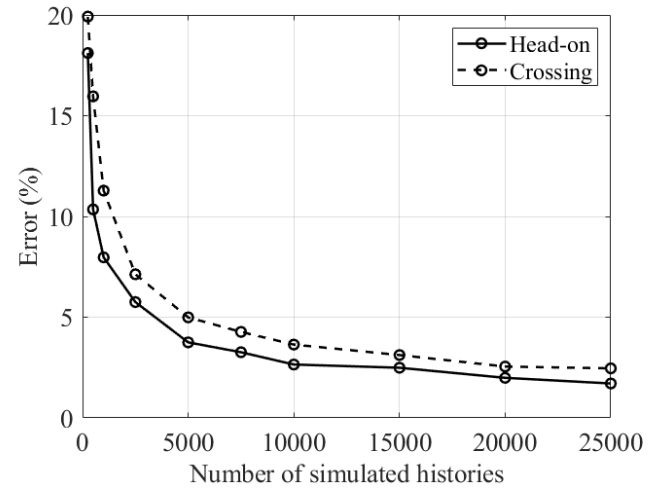


Figure 5. Error related to the 95% CI of the probability of collision for different number of simulated histories.

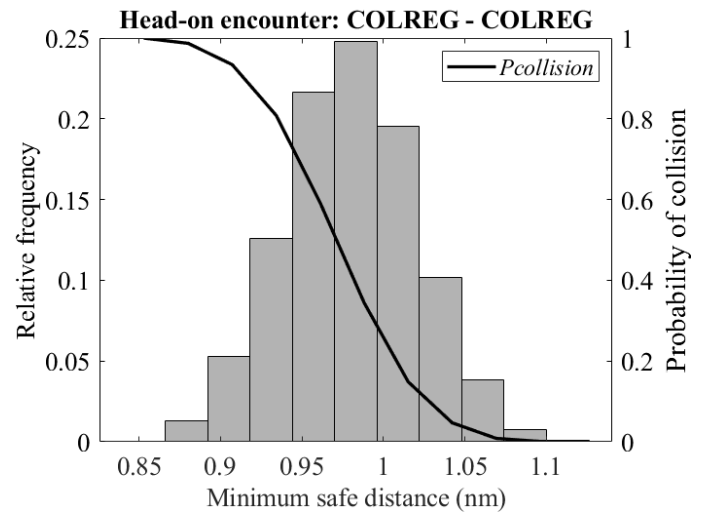


Figure 6. Histogram of minimum safe distance in nautical miles with respective probability distribution function of collision.

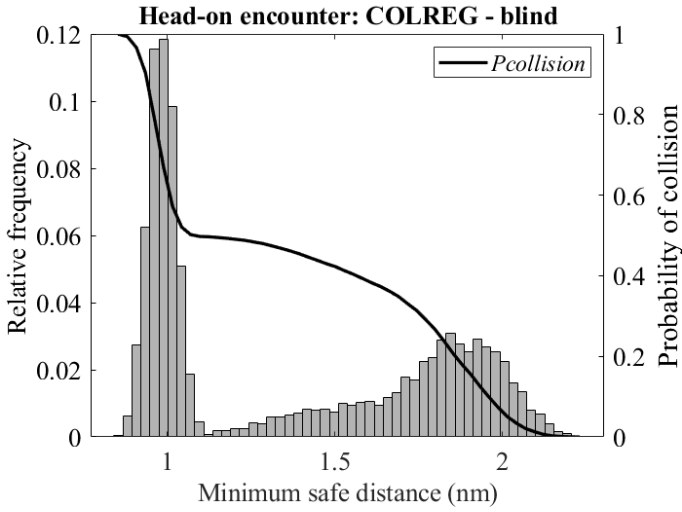


Figure 7. Histogram of minimum safe distance in nautical miles with respective probability distribution function of collision.

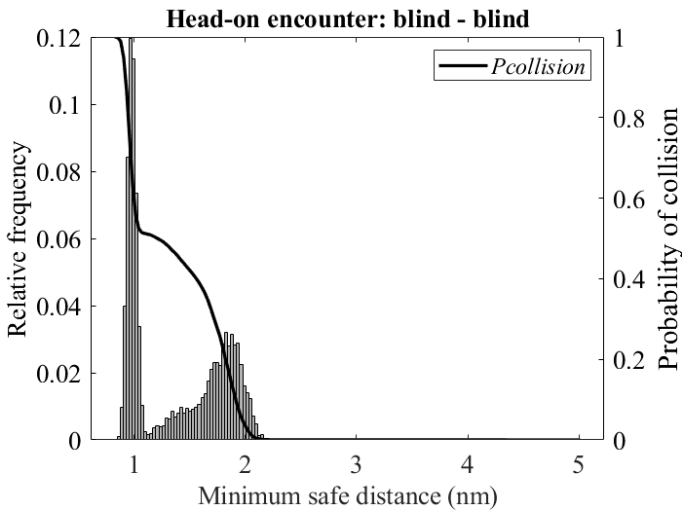


Figure 8. Histogram of minimum safe distance in nautical miles with respective probability distribution function of collision.

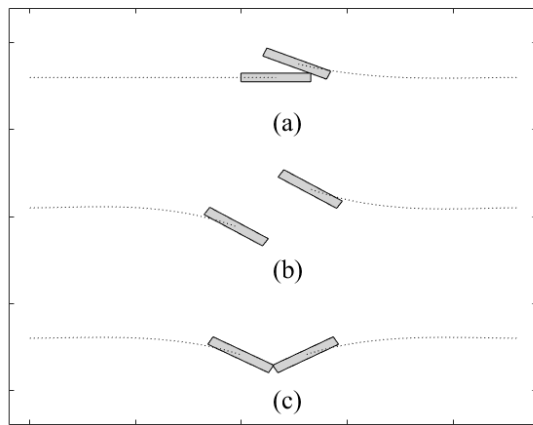


Figure 9. Different behaviours of vessels in close quarter situation during a head-on encounter.

Figure 7 illustrates the minimum safe distance histogram, along with the associated probability distribution function for collision events in the COLREG-blind scenario. The results indicate that when only one vessel is adhering to the COLREG rules while the other has a random rudder behaviour, the probability of collision increases as the distance between the

vessels decreases below 2.5 nm. When the distance between the vessels is less than 1.1 nm, the probability of collision exceeds 60%.

Figure 8 depicts the minimum safe distance histogram, along with the corresponding probability distribution function for collision events in the blind-blind scenario. The results reveal that the probability distribution function of collision in this scenario is very similar to that observed in the COLREG-blind case. Therefore, in close-quarter situations, the probability of collision in head-on encounters is roughly the same whether only one of the vessels follows the rules or neither vessel follows the rules.

The probability of collision being similar is not the only similarity between Figure 7 and Figure 8. Both cases result in bimodal histograms representing the presence of two subpopulations in the sample. This occurs due to different possible manoeuvre combinations between the vessels, as shown in Figure 9. The first peak in both figures corresponds to the minimum safe distance necessary to avoid frontal collisions (see Figure 9a). The histograms of Figure 7 and Figure 8 also show a relative frequency valley between peaks, which corresponds to the situations when both vessels pass parallel to each other. In this case, the probability distribution function stabilises its increasing because, despite a possibility of side-to-side collision, the vessels can pass each other without colliding as shown in Figure 9b. The second histogram peak, at a higher distance, corresponds to simulations when both vessels are away from each other, but both apply the rudder to the same geographic direction (see Figure 9c).

4.4 Probability of collision in crossing situation

Considering manoeuvring characteristics of the vessel and applying the optimum number of simulation histories, this section presents a preliminary risk assessment for collisions in crossing situation.

Figure 10 shows the histogram of the minimum safe distance, along with the corresponding probability distribution function for collision events in the COLREG-COLREG scenario. The findings demonstrate that if both vessels comply with the rules specified in COLREG, the probability of collision is minimal for distances up to 1.5 nm between the vessels, which is slightly higher compared to the head-on.

However, for the type of vessel considered in this study, collisions are unavoidable when the distance between vessels is less than 0.5 nm, even if both vessels are following the COLREG rules.

Figure 11 presents the histogram of the minimum safe distance along with the respective probability distribution function when only one of the vessels follows the COLREG rules, while the other has a random behaviour of rudder. Besides, Figure 12 displays similar results to Figure 11, but in this case, both vessels can apply any rudder (i.e. the blind-blind scenario).

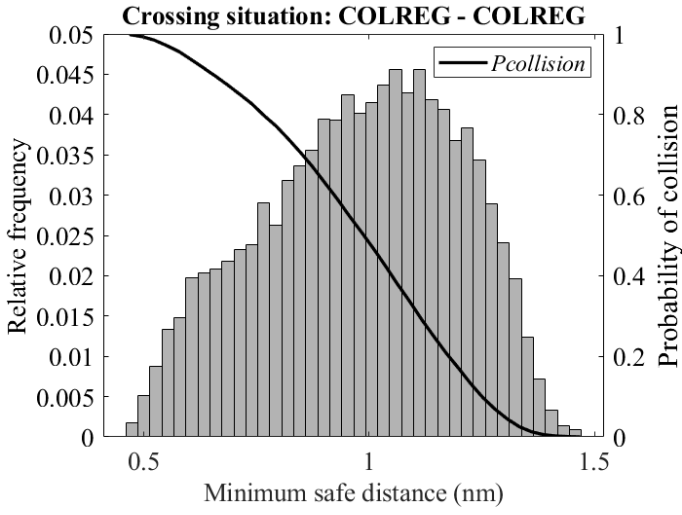


Figure 10. Histogram of minimum safe distance in nautical miles with respective probability distribution function of collision.

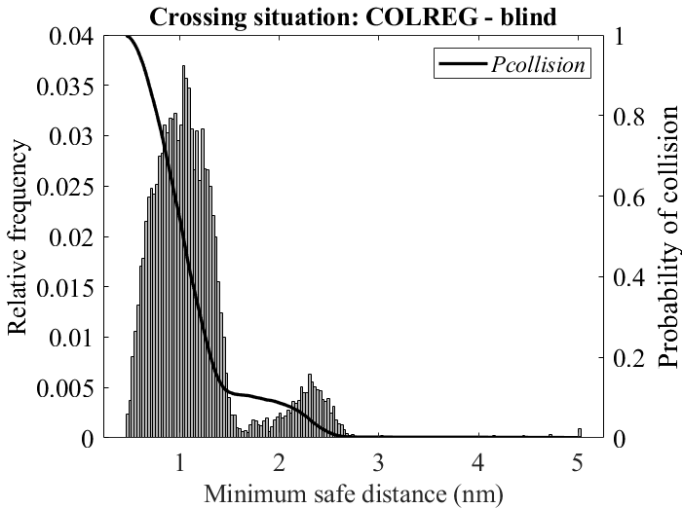


Figure 11. Histogram of minimum safe distance in nautical miles with respective probability distribution function of collision.

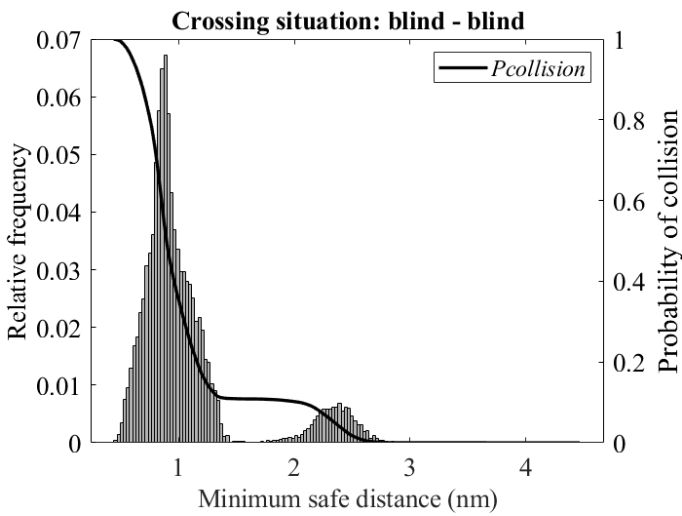


Figure 12. Histogram of minimum safe distance in nautical miles with respective probability distribution function of collision.

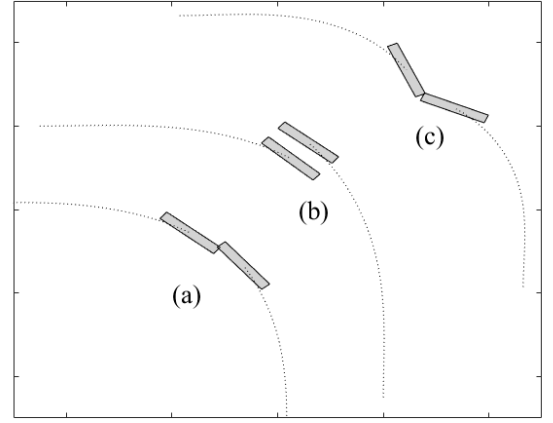


Figure 13. Different behaviours of vessels in close quarter situation during a crossing encounter.

In crossing situations, similar to the head-on encounter scenario, the histograms of Figure 11 and Figure 12 exhibit bimodal behaviour. The first peak in both figures corresponds to the minimum safe distance necessary to avoid frontal collisions (as shown in Figure 13a). The relative frequency valley between peaks represents situations where both vessels pass parallel to each other. Here, the probability distribution function stabilises its increase because, despite the possibility of a side-to-side collision, the vessels can pass each other without colliding, as illustrated in Figure 13b. The second peak of the histograms corresponds to simulations where both vessels are far away from each other but apply the rudder to the same geographic direction (as shown in Figure 13c).

5 CONCLUSIONS

This paper presents an initial assessment of the collision probability in head-on encounter and crossing situation using a trajectory simulation of two cargo ships with identical dimensions. The assessment is based on the concept of minimum safe distance, which considers the manoeuvrability characteristics, initial speed, main dimensions, and vessel type. The results are derived using the first-order Nomoto model and Monte Carlo Simulation techniques. Three distinct scenarios are employed for both head-on and crossing collision encounters: COLREG-COLREG, COLREG-blind, and blind-blind. Based on the simulation results, the probability distribution function of collision is derived for each scenario. The proposed model provides a preliminary risk assessment of collision in head-on encounters and crossing situations that can be improved and developed further for collision avoidance applications. Overall, this paper demonstrates the importance of considering the manoeuvring capabilities of vessels and minimum safe distance concept in collision risk assessment.

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